

# DATA\_Final\_Proj\_LogisticReg

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2026-05-09

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.6.0
## v ggplot2     4.0.0      v tibble     3.2.1
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.2.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
spotify <- read.csv("spotify.csv")
```

```
dim(spotify)
```

```
## [1] 114000      21
```

```
glimpse(spotify)
```

```
## Rows: 114,000
## Columns: 21
## $ X               <int> 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, ~
## $ track_id        <chr> "5Su0ikwiRyPMVoIQDJUGSV", "4qPNDBW1i3p13qLCtOKi3A", "~
## $ artists         <chr> "Gen Hoshino", "Ben Woodward", "Ingrid Michaelson;ZAY~
## $ album_name       <chr> "Comedy", "Ghost (Acoustic)", "To Begin Again", "Craz~
## $ track_name       <chr> "Comedy", "Ghost - Acoustic", "To Begin Again", "Can'~
## $ popularity       <int> 73, 55, 57, 71, 82, 58, 74, 80, 74, 56, 74, 69, 52, 6~
## $ duration_ms      <int> 230666, 149610, 210826, 201933, 198853, 214240, 22940~
## $ explicit         <chr> "False", "False", "False", "False", "False", "False", ~
## $ danceability      <dbl> 0.676, 0.420, 0.438, 0.266, 0.618, 0.688, 0.407, 0.70~
## $ energy            <dbl> 0.4610, 0.1660, 0.3590, 0.0596, 0.4430, 0.4810, 0.147~
## $ key               <int> 1, 1, 0, 0, 2, 6, 2, 11, 0, 1, 8, 4, 7, 3, 2, 4, 2, 1~
## $ loudness          <dbl> -6.746, -17.235, -9.734, -18.515, -9.681, -8.807, -8.~
## $ mode              <int> 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 0, ~
## $ speechiness       <dbl> 0.1430, 0.0763, 0.0557, 0.0363, 0.0526, 0.1050, 0.035~
## $ acousticness      <dbl> 0.0322, 0.9240, 0.2100, 0.9050, 0.4690, 0.2890, 0.857~
## $ instrumentalness  <dbl> 1.01e-06, 5.56e-06, 0.00e+00, 7.07e-05, 0.00e+00, 0.0~
## $ liveness          <dbl> 0.3580, 0.1010, 0.1170, 0.1320, 0.0829, 0.1890, 0.091~
## $ valence           <dbl> 0.7150, 0.2670, 0.1200, 0.1430, 0.1670, 0.6660, 0.076~
## $ tempo             <dbl> 87.917, 77.489, 76.332, 181.740, 119.949, 98.017, 141~
## $ time_signature    <int> 4, 4, 4, 3, 4, 4, 3, 4, 4, 4, 4, 3, 4, 4, 4, 3, 4, 4, ~
## $ track_genre       <chr> "acoustic", "acoustic", "acoustic", "acoustic", "acou~
```

```
summary(spotify)
```

```
##           X           track_id           artists           album_name
## Min.      :    0   Length:114000   Length:114000   Length:114000
## 1st Qu.: 28500   Class :character   Class :character   Class :character
## Median : 57000   Mode  :character   Mode  :character   Mode  :character
## Mean      : 57000
## 3rd Qu.: 85499
## Max.      :113999
##   track_name      popularity      duration_ms      explicit
## Length:114000    Min.      :  0.00   Min.      :    0   Length:114000
## Class :character 1st Qu.: 17.00   1st Qu.: 174066   Class :character
## Mode  :character Median : 35.00   Median : 212906   Mode  :character
##                  Mean      : 33.24   Mean      : 228029
##                  3rd Qu.: 50.00   3rd Qu.: 261506
##                  Max.      :100.00   Max.      :5237295
##   danceability      energy      key      loudness
## Min.      :0.0000   Min.      :0.0000   Min.      : 0.000   Min.      : -49.531
## 1st Qu.:0.4560   1st Qu.:0.4720   1st Qu.:  2.000   1st Qu.: -10.013
## Median :0.5800   Median :0.6850   Median :  5.000   Median :  -7.004
## Mean      :0.5668   Mean      :0.6414   Mean      : 5.309   Mean      : -8.259
## 3rd Qu.:0.6950   3rd Qu.:0.8540   3rd Qu.:  8.000   3rd Qu.:  -5.003
## Max.      :0.9850   Max.      :1.0000   Max.      :11.000   Max.      :  4.532
##      mode      speechiness      acousticness      instrumentalness
## Min.      :0.0000   Min.      :0.00000   Min.      :0.00000   Min.      :0.00e+00
## 1st Qu.:0.0000   1st Qu.:0.03590   1st Qu.:0.0169   1st Qu.:0.00e+00
## Median :1.0000   Median :0.04890   Median :0.1690   Median :4.16e-05
## Mean      :0.6376   Mean      :0.08465   Mean      :0.3149   Mean      :1.56e-01
## 3rd Qu.:1.0000   3rd Qu.:0.08450   3rd Qu.:0.5980   3rd Qu.:4.90e-02
## Max.      :1.0000   Max.      :0.96500   Max.      :0.9960   Max.      :1.00e+00
##   liveness      valence      tempo      time_signature
## Min.      :0.0000   Min.      :0.0000   Min.      :  0.00   Min.      :0.000
## 1st Qu.:0.0980   1st Qu.:0.2600   1st Qu.: 99.22   1st Qu.:4.000
## Median :0.1320   Median :0.4640   Median :122.02   Median :4.000
## Mean      :0.2136   Mean      :0.4741   Mean      :122.15   Mean      :3.904
## 3rd Qu.:0.2730   3rd Qu.:0.6830   3rd Qu.:140.07   3rd Qu.:4.000
## Max.      :1.0000   Max.      :0.9950   Max.      :243.37   Max.      :5.000
##   track_genre
## Length:114000
## Class :character
## Mode  :character
##
##
##
```

```
# cleaning
spotify <- spotify %>%
  distinct() %>%
  mutate(
    explicit = as.factor(explicit),
    track_genre = as.factor(track_genre),
    duration_min = duration_ms / 60000 )

spotify <- spotify %>%
```

```

mutate(
  track_genre = case_when(
    track_genre %in% c("metal", "metalcore") ~ "metal",
    track_genre %in% c("deep-house", "electronic", "progressive-house") ~ "electronic_house",
    track_genre %in% c("minimal-techno", "techno") ~ "techno",
    track_genre %in% c("pop", "pop-film") ~ "pop",
    TRUE ~ as.character(track_genre)))

# 20 most popular genres
top_20_popular <- spotify %>%
  group_by(track_genre) %>%
  summarise(avg_popularity = mean(popularity, na.rm = TRUE)) %>%
  arrange(desc(avg_popularity)) %>%
  slice_head(n = 20) %>%
  pull(track_genre)

spotify_top20 <- spotify %>%
  filter(track_genre %in% top_20_popular) %>%
  mutate(track_genre = factor(track_genre))

summary(spotify)

```

```

##           X           track_id           artists           album_name
## Min.      :    0   Length:114000   Length:114000   Length:114000
## 1st Qu.: 28500   Class :character   Class :character   Class :character
## Median : 57000   Mode  :character   Mode  :character   Mode  :character
## Mean      : 57000
## 3rd Qu.: 85499
## Max.      :113999
##   track_name      popularity      duration_ms      explicit
## Length:114000   Min.      : 0.00   Min.      :    0   False:104253
## Class :character 1st Qu.: 17.00   1st Qu.: 174066   True  :  9747
## Mode  :character Median : 35.00   Median : 212906
##                      Mean  : 33.24   Mean  : 228029
##                      3rd Qu.: 50.00   3rd Qu.: 261506
##                      Max.    :100.00   Max.    :5237295
##   danceability      energy      key      loudness
## Min.      :0.0000   Min.      :0.0000   Min.      : 0.000   Min.      : -49.531
## 1st Qu.:0.4560   1st Qu.:0.4720   1st Qu.: 2.000   1st Qu.: -10.013
## Median :0.5800   Median :0.6850   Median : 5.000   Median :  -7.004
## Mean      :0.5668   Mean      :0.6414   Mean      : 5.309   Mean      : -8.259
## 3rd Qu.:0.6950   3rd Qu.:0.8540   3rd Qu.: 8.000   3rd Qu.:  -5.003
## Max.      :0.9850   Max.      :1.0000   Max.      :11.000   Max.      :  4.532
##      mode      speechiness      acousticness      instrumentalness
## Min.      :0.0000   Min.      :0.00000   Min.      :0.0000   Min.      :0.00e+00
## 1st Qu.:0.0000   1st Qu.:0.03590   1st Qu.:0.0169   1st Qu.:0.00e+00
## Median :1.0000   Median :0.04890   Median :0.1690   Median :4.16e-05
## Mean      :0.6376   Mean      :0.08465   Mean      :0.3149   Mean      :1.56e-01
## 3rd Qu.:1.0000   3rd Qu.:0.08450   3rd Qu.:0.5980   3rd Qu.:4.90e-02
## Max.      :1.0000   Max.      :0.96500   Max.      :0.9960   Max.      :1.00e+00
##   liveness      valence      tempo      time_signature
## Min.      :0.0000   Min.      :0.0000   Min.      : 0.00   Min.      :0.000
## 1st Qu.:0.0980   1st Qu.:0.2600   1st Qu.: 99.22   1st Qu.:4.000
## Median :0.1320   Median :0.4640   Median :122.02   Median :4.000

```

```
## Mean :0.2136 Mean :0.4741 Mean :122.15 Mean :3.904
## 3rd Qu.:0.2730 3rd Qu.:0.6830 3rd Qu.:140.07 3rd Qu.:4.000
## Max. :1.0000 Max. :0.9950 Max. :243.37 Max. :5.000
## track_genre duration_min
## Length:114000 Min. : 0.000
## Class :character 1st Qu.: 2.901
## Mode :character Median : 3.548
## Mean : 3.800
## 3rd Qu.: 4.358
## Max. :87.288
```

```
# checking for distinctness in genres
top20_summary <- spotify_top20 %>%
  group_by(track_genre) %>%
  summarise(
    danceability = mean(danceability),
    energy = mean(energy),
    acousticness = mean(acousticness),
    speechiness = mean(speechiness),
    instrumentalness = mean(instrumentalness),
    valence = mean(valence),
    tempo = mean(tempo)
  )

top20_summary
```

```
## # A tibble: 20 x 8
##   track_genre danceability energy acousticness speechiness instrumentalness
##   <fct>          <dbl> <dbl>         <dbl>         <dbl>         <dbl>
## 1 acoustic      0.550 0.435         0.567         0.0432        0.0383
## 2 ambient       0.368 0.237         0.777         0.0416        0.676
## 3 anime         0.537 0.674         0.268         0.0873        0.263
## 4 brazil        0.563 0.621         0.332         0.0960        0.0131
## 5 british       0.501 0.507         0.428         0.0500        0.194
## 6 chill         0.664 0.427         0.534         0.102         0.178
## 7 electronic_hou~ 0.662 0.750         0.113         0.0698        0.229
## 8 emo           0.599 0.670         0.195         0.109         0.0303
## 9 grunge        0.457 0.803         0.0534        0.0611        0.0370
## 10 hard-rock     0.482 0.795         0.0904        0.0665        0.0496
## 11 indian        0.592 0.567         0.485         0.0721        0.0394
## 12 k-pop         0.648 0.676         0.293         0.0880        0.00983
## 13 mandopop     0.547 0.498         0.480         0.0395        0.00371
## 14 metal         0.444 0.877         0.0237        0.0903        0.0551
## 15 pagode       0.578 0.712         0.538         0.0628        0.000566
## 16 piano        0.455 0.320         0.705         0.0469        0.516
## 17 pop          0.614 0.605         0.392         0.0720        0.00858
## 18 psych-rock    0.503 0.562         0.315         0.0473        0.0918
## 19 sad          0.692 0.462         0.474         0.130         0.106
## 20 sertanejo    0.592 0.710         0.437         0.0649        0.000160
## # i 2 more variables: valence <dbl>, tempo <dbl>
```

```
# multinomial logistic regression
library(nnet)

set.seed(1)
```

```

train_index <- sample(1:nrow(spotify_top20), 0.7 * nrow(spotify_top20))

train_data <- spotify_top20[train_index, ]
test_data <- spotify_top20[-train_index, ]

genre_model_top20 <- multinom(
  track_genre ~ popularity + explicit + danceability + energy + key + mode + speechiness + acousticness
  data = train_data,
  MaxNWts = 5000,
  maxit = 500
)

```

```

## # weights: 320 (285 variable)
## initial value 50328.302196
## iter 10 value 49165.662895
## iter 20 value 47578.332823
## iter 30 value 45937.772370
## iter 40 value 43176.210006
## iter 50 value 40809.417426
## iter 60 value 39294.601216
## iter 70 value 38296.104804
## iter 80 value 37354.101905
## iter 90 value 36263.950863
## iter 100 value 34985.290211
## iter 110 value 34553.990029
## iter 120 value 34357.230720
## iter 130 value 34262.241038
## iter 140 value 34162.848363
## iter 150 value 34111.678730
## iter 160 value 34073.664332
## iter 170 value 34052.838531
## iter 180 value 34038.155889
## iter 190 value 34031.537802
## iter 200 value 34025.342410
## iter 210 value 34021.291536
## iter 220 value 34017.212987
## iter 230 value 34011.457216
## iter 240 value 34004.986549
## iter 250 value 34001.178396
## iter 260 value 33997.646835
## iter 270 value 33993.441276
## iter 280 value 33992.313419
## iter 290 value 33992.208887
## iter 290 value 33992.208697
## iter 290 value 33992.208603
## final value 33992.208603
## converged

```

```

# Predictability
pred_top20 <- predict(genre_model_top20, newdata = test_data)

mean(pred_top20 == test_data$track_genre)

```

```
## [1] 0.3569444
```

```

# Odds ratios
odds_table_top20 <- as.data.frame(exp(coef(genre_model_top20)))

odds_table_top20$genre <- rownames(odds_table_top20)

odds_table_top20 <- odds_table_top20 %>%
  select(genre, everything())

odds_long_top20 <- odds_table_top20 %>%
  pivot_longer(cols = -genre, names_to = "characteristic", values_to = "odds_ratio") %>%
  filter(characteristic != "(Intercept)") %>%
  mutate(effect_strength = abs(log(odds_ratio)))

odds_long_top20 %>%
  group_by(characteristic) %>%
  summarise(avg_strength = mean(effect_strength)) %>%
  arrange(desc(avg_strength))

## # A tibble: 14 x 2
##   characteristic    avg_strength
##   <chr>              <dbl>
## 1 instrumentality    10.6
## 2 speechiness        8.15
## 3 energy             3.83
## 4 danceability       3.79
## 5 acousticness       2.41
## 6 valence            2.00
## 7 liveness           1.66
## 8 explicitTrue       1.04
## 9 mode               0.700
## 10 time_signature    0.389
## 11 duration_min      0.342
## 12 popularity        0.0162
## 13 key               0.0112
## 14 tempo             0.00365

# Pseudo R squared
null_model_top20 <- multinom(track_genre ~ 1, data = train_data, trace = FALSE)

1 - (deviance(genre_model_top20) / deviance(null_model_top20))

## [1] 0.3080694

# Confusion Matrix
library(caret)

## Loading required package: lattice

##
## Attaching package: 'caret'

## The following object is masked from 'package:purrr':
##
##   lift

pred_top20 <- factor(pred_top20, levels = levels(test_data$track_genre))

```

```
cm_top20 <- confusionMatrix(
  data = pred_top20,
  reference = test_data$track_genre
)
cm_top20
```

```
## Confusion Matrix and Statistics
```

```
##
```

```
##
##               Reference
## Prediction      acoustic ambient anime brazil british chill electronic_house
## acoustic          78      17      1      7      33      15          1
## ambient            3     169     28      0      30      7          9
## anime              0      3      8      2      3      6         14
## brazil             3      3      7     42      9      1          7
## british            1      0      3      0      3      2          1
## chill             20      5     22      1      6     72          7
## electronic_house   22     18     71     30     28     39         668
## emo                7      0      8      4      6     15         27
## grunge             6      4      7      7      6      0         10
## hard-rock          17      2     14     11      9      0          8
## indian             0      0      3      0      2      2          3
## k-pop              0      0      1      4      0      1          4
## mandopop           52     15      4     42     18      6          9
## metal              12      4     73     19     15      2         53
## pagode             8      0      4     13      5      1          1
## piano              5     47     16      1     18      8          5
## pop                34      3     19     49     42     44         45
## psych-rock         14      1      5     19     45      4         10
## sad                9      2      8     13      3     59          4
## sertanejo          6      1      6     21     14     10         12
```

```
##
##               Reference
## Prediction      emo grunge hard-rock indian k-pop mandopop metal pagode
## acoustic          3      3          5     42      8         39      5      8
## ambient            2      1          1      2      0          1      2      0
## anime              0      2          6      2      4          0      4      0
## brazil             2      2          2      7      9          4      3      5
## british            1      0          1      3      0          0      2      0
## chill              9      1          0     13      2          1      0      0
## electronic_house   45     38         30     49    107         47     72     17
## emo                55      2          0      5      0          2      7      3
## grunge             7     38         23      2      3          3     14      0
## hard-rock          17     18         40      6      3          3     24      1
## indian             2      1          1      4      3          0      0      0
## k-pop              0      0          2      3     13          0      1      0
## mandopop           2      7         11     23     10         94      7      1
## metal              66    183        116      3      9          6    442      1
## pagode             2      0          6      4      9          4      1    182
## piano              0      0          1     10      0          0      1      0
## pop                13     14         32    128    104         72      8     36
## psych-rock         6      8         34      5      6         14     11      5
## sad                61      0          0      8      3          0      0      2
## sertanejo          2      8          8      5      9          3      2     50
```

```
##
##               Reference
```

```
## Prediction      piano pop psych-rock sad sertanejo
```

```

## acoustic      19 21      21 11      8
## ambient      82 2       9 2       0
## anime        1 1       3 7       0
## brazil       1 8       13 2      6
## british      8 5       0 1       0
## chill        8 12      8 50      0
## electronic_house 28 109    20 36     21
## emo          2 19      2 50      0
## grunge       8 7       6 1       1
## hard-rock    7 10      18 0       2
## indian       0 1       0 0       0
## k-pop        1 26      1 0       0
## mandopop     14 24     4 4       6
## metal        10 6      26 0       2
## pagode       4 20     12 0      63
## piano        77 2      2 3       0
## pop          24 268    37 15     33
## psych-rock   10 14     84 1      14
## sad          0 12      3 103     1
## sertanejo    5 9       20 6     130
##
## Overall Statistics
##
## Accuracy : 0.3569
## 95% CI : (0.3459, 0.3681)
## No Information Rate : 0.1247
## P-Value [Acc > NIR] : < 2.2e-16
##
## Kappa : 0.3082
##
## McNemar's Test P-Value : NA
##
## Statistics by Class:
##
## Class: acoustic Class: ambient Class: anime Class: brazil
## Sensitivity      0.26263      0.57483      0.025974      0.147368
## Specificity      0.96132      0.97379      0.991584      0.986406
## Pos Pred Value    0.22609      0.48286      0.121212      0.308824
## Neg Pred Value    0.96805      0.98175      0.957948      0.965600
## Prevalence       0.04125      0.04083      0.042778      0.039583
## Detection Rate    0.01083      0.02347      0.001111      0.005833
## Detection Prevalence 0.04792      0.04861      0.009167      0.018889
## Balanced Accuracy 0.61197      0.77431      0.508779      0.566887
##
## Class: british Class: chill Class: electronic_house
## Sensitivity      0.0101695      0.24490      0.74388
## Specificity      0.9959450      0.97611      0.86877
## Pos Pred Value    0.0967742      0.30380      0.44682
## Neg Pred Value    0.9592691      0.96812      0.95968
## Prevalence       0.0409722      0.04083      0.12472
## Detection Rate    0.0004167      0.01000      0.09278
## Detection Prevalence 0.0043056      0.03292      0.20764
## Balanced Accuracy 0.5030572      0.61050      0.80632
##
## Class: emo Class: grunge Class: hard-rock Class: indian
## Sensitivity      0.186441      0.116564      0.125392      0.0123457

```

```

## Specificity          0.976973      0.983270      0.975294      0.9973822
## Pos Pred Value      0.257009      0.248366      0.190476      0.1818182
## Neg Pred Value      0.965646      0.959132      0.960086      0.9554193
## Prevalence          0.040972      0.045278      0.044306      0.0450000
## Detection Rate      0.007639      0.005278      0.005556      0.0005556
## Detection Prevalence 0.029722      0.021250      0.029167      0.0030556
## Balanced Accuracy    0.581707      0.549917      0.550343      0.5048639
##
## Class: k-pop Class: mandopop Class: metal Class: pagode
## Sensitivity          0.043046      0.32082      0.72937      0.58521
## Specificity          0.993621      0.96250      0.90810      0.97721
## Pos Pred Value      0.228070      0.26629      0.42176      0.53687
## Neg Pred Value      0.959541      0.97094      0.97334      0.98120
## Prevalence          0.041944      0.04069      0.08417      0.04319
## Detection Rate      0.001806      0.01306      0.06139      0.02528
## Detection Prevalence 0.007917      0.04903      0.14556      0.04708
## Balanced Accuracy    0.518334      0.64166      0.81874      0.78121
##
## Class: piano Class: pop Class: psych-rock Class: sad
## Sensitivity          0.24919      0.46528      0.29066      0.35274
## Specificity          0.98273      0.88647      0.96730      0.97279
## Pos Pred Value      0.39286      0.26275      0.27097      0.35395
## Neg Pred Value      0.96688      0.95016      0.97025      0.97264
## Prevalence          0.04292      0.08000      0.04014      0.04056
## Detection Rate      0.01069      0.03722      0.01167      0.01431
## Detection Prevalence 0.02722      0.14167      0.04306      0.04042
## Balanced Accuracy    0.61596      0.67588      0.62898      0.66276
##
## Class: sertanejo
## Sensitivity          0.45296
## Specificity          0.97150
## Pos Pred Value      0.39755
## Neg Pred Value      0.97716
## Prevalence          0.03986
## Detection Rate      0.01806
## Detection Prevalence 0.04542
## Balanced Accuracy    0.71223

```

```

# Precision and F1
metrics_top20 <- as.data.frame(cm_top20$byClass)

metrics_top20 %>%
  summarise(avg_precision = mean(Precision, na.rm = TRUE), avg_f1 = mean(F1, na.rm = TRUE))

```

```

## avg_precision avg_f1
## 1 0.2997554 0.2781557

```

This model used multinomial logistic regression to classify spotify songs into the 20 genres with the highest average popularity (24,000 songs). Similar genres were combined prior to modeling to reduce overlap. Instrumentalness, speechiness, energy, danceability, and acousticness are the top 5 strongest predictors of genre. Audio texture and style features are more useful for genre classification than musical attributes like key, tempo, or popularity.

The model achieved: - 35.69% accuracy - 29.9976% precision - 27.82% F1 Score - .308 pseudo R squared

Genres such as metal, electronic house, ambient, and sertanejo were easier to classify because they have more distinct audio features. Broader genres like pop were more difficult to distinguish.